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Unsupervised Disambiguation for a Multilingual Medical Information System using UMLS

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Abstract

This paper describes techniques for unsupervised word sense disambiguation of English and German medical documents using the Unified Medical Language System (UMLS). We present both monolingual techniques which rely only on the structure of UMLS, and bilingual techniques which also rely on the availability of parallel corpora. The best results are obtained using relationships between terms given by UMLS, a method which achieves 74% precision, 66% coverage for English and 79% precision, 73% coverage for German on evaluation corpora and over 83% coverage over the whole corpus. The success of this technique for German shows that a lexical resource giving relationships between concepts used to index an English document collection can be used for high quality disambiguation in another language.

Document retrieval experiments comparing models using semantic annotation and disambiguation show that a search engine which retrieves documents using the semantic annotation with UMLS concepts is much more effective than a baseline search engine which indexes only tokens. In addition, adding the automatic disambiguation component to this search engine gave a small but reliable improvement in precision at high-recall document retrieval.

Key words: Unsupervised Disambiguation, Cross-Lingual Information Retrieval

1 Introduction

This paper reports on experiments in monolingual and multilingual word sense disambiguation (WSD) in the medical domain using the Unified Medical Language System (UMLS). The work described was carried out as part of the

MUCHMORE project¹ for multilingual organisation and retrieval of medical information, for which WSD is particularly important.

The importance of WSD to multilingual applications stems from the simple fact that meanings represented by a single word in one language may be represented by multiple words in other languages. The English word *drug* when referring to medically therapeutic drugs would be translated as *medikamente*, while it would be rendered as *drogen* when referring to a recreationally taken narcotic substance of the kind that many governments prohibit by law.

The ability to disambiguate is therefore essential to the task of machine translation — when translating from English to Spanish or from English to German we would need to make the distinctions mentioned above and other similar ones. Even short of the task of full translation, WSD should be crucial to improving applications such as cross-lingual information retrieval (CLIR), since search terms entered in the language used for querying must be appropriately rendered in the language used for retrieval.

Methods for WSD can effectively be divided into those that require manually annotated training data (supervised methods) and those that do not (unsupervised methods) (Ide and Véronis, 1998). In general, supervised methods are less scalable than unsupervised methods because they rely on training data which may be costly and unrealistic to produce, and even then might be available for only a few ambiguous terms. The goal of our work on disambiguation in the MUCHMORE project is to enable the correct semantic annotation of entire document collections with all terms which are potentially relevant for organisation, retrieval and summarisation of information (full ‘Sense Tagging’ as defined by Stevenson (2003, Ch 1)), the lexicon of senses being provided by UMLS). Therefore a decision was taken early on in the project that we should focus on unsupervised methods, which have the potential to be scaled up enough to meet our needs. Instead of using methods which rely on manually annotated training examples, this need for scalability encouraged us to look at other resources which were already available, including monolingual and bilingual corpora, knowledge bases and lexical resources such as GermaNet (for general word senses) and UMLS (for the medical domain).

As with many ‘unsupervised’ methods for disambiguation, these methods still rely heavily on external resources, but upon resources for which large amounts of investment are readily provided independently of any potential uses for word sense disambiguation. For example, one method for disambiguation makes use of the work that has gone into using UMLS concepts as a controlled vocabulary for indexing the MEDLINE library, a process which is carried out by medical experts. Thus a great amount of supervision has gone into creating

¹ <http://muchmore.dfki.de>

this knowledge source: nonetheless, for use in word sense disambiguation, this information comes to us for free. The same point could be made of any dictionary based disambiguation system, such as the traditional Lesk method (Lesk (1986)).

As well as the two methods described in this paper, automatic disambiguation methods developed by the MUCHMORE project and described elsewhere include:

- Collocational disambiguation, where the meaning of an individual ambiguous word (such as *vessel*) can be determined by its occurrence in a fixed expression such as *blood vessel* (Widdows et al., 2003a).
- Domain-specific sense selection (Buitelaar and Sacaleanu, 2001), automatically selects senses (i.e. synsets) on the basis of the relevance of their constituent terms that co-occur within representative domain corpora.
- Instance-based learning implements a k-nearest neighbor algorithm, in which sense disambiguation is treated as a classification task, i.e. an ambiguous word will be classified to the appropriate class (sense) given a particular context (Steffen et al., 2003).

The rest of this paper is arranged as follows. In Section 2 we describe the lexical resource (UMLS) and the corpora we used for our experiments. We then describe and evaluate two different methods for disambiguation. The bilingual method (Section 3) takes advantage of our having a translated corpus, because knowing the translation of an ambiguous word can be enough to determine its sense. UMLS relation based methods (Section 4) use relationships between terms in UMLS to determine which sense is being used in a particular instance.

The method based on UMLS relationships, a descendent of Lesk’s early dictionary based method (Lesk, 1986), achieved excellent precision and coverage on both English and German corpora. It was therefore used as a component in a cross lingual search engine, and gave modest but reliable improvements in the precision of high-recall document retrieval. These results are described in Section 5.

2 Language resources used in these experiments

2.1 Lexical Resource — UMLS

The Unified Medical Language System (UMLS) is a resource that contains linguistic, terminological and semantic information in the medical domain.² It is organised in three parts: Specialist Lexicon, MetaThesaurus and Semantic Network. The MetaThesaurus contains concepts from more than 60 standardised medical thesauri, of which for our purposes we only use the concepts from MeSH (the Medical Subject Headings thesaurus). This decision is based on the fact that MeSH is also available in German. The semantic information that we use in annotation is the so-called Concept Unique Identifier (CUI), a code that represents a concept in the UMLS MetaThesaurus. We consider the possible ‘senses’ of a term to be the set of CUI’s which list this term as a possible realisation. For example, UMLS contains the term *trauma* as a possible realisation of the following two concepts:³

C0043251 Injuries and Wounds: Wounds and Injuries: trauma: traumatic disorders: Traumatic injury:

C0021501 Physical Trauma: Trauma (Physical): trauma:

Each of these CUI’s is a possible sense of the term *trauma*. The term *trauma* is therefore noted as ambiguous, since it can be used to express more than one UMLS concept. The purpose of disambiguation is to find out which of these possible senses is actually being used in each particular context where the term *trauma* is used.

CUI’s in UMLS are also interlinked to each other by a number of thesaural relationships.⁴ These include:

- ‘Broader term’ which is similar to the hypernymy relation in WordNet (Fellbaum, 1998). In general, x is a ‘broader term’ for y if every y is also a (kind of) x .

² UMLS is freely available under license from the United States National Library of Medicine, <http://www.nlm.nih.gov/research/umls/>

³ Each of the separate terms in these list can be a realisation of the concept which is defined by their ‘intersection’: such a list of terms is thus equivalent to the notion of a ‘synset’ in WordNet (Fellbaum, 1998).

⁴ These thesaural relationships should be distinguished from the *semantic relations* of the UMLS *semantic network*, which are between general semantic types for example “*drug treats disease*” rather than specific instances. Recognising examples of these general semantic network relations in particular documents can also be used to improve information retrieval (Vintar et al., 2003).

- More generally, ‘related terms’ are listed, relationships including ‘is like’, ‘is clinically associated with’.
- Cooccurring concepts, which are pairs of concepts which are linked in some information source. In particular, two concepts are regarded as cooccurring if they have both been used to manually index the same document in MEDLINE. We will refer to such pairs of concepts as *coindexing* concepts.

2.2 The Springer Corpus of Medical Abstracts

The experiments and implementations of WSD described in this paper were all carried out on a parallel corpus of English-German medical scientific abstracts obtained from the Springer Link web site.⁵ The corpus consists of approximately 1 million tokens for each language. Abstracts are from 41 medical journals, each of which constitutes a relatively homogeneous medical sub-domain (e.g. Neurology, Radiology, etc.). The corpus was automatically marked up with morphosyntactic and semantic information, as described in (Vintar et al., 2002). In brief, whenever a token is encountered in the corpus that is listed as a term in UMLS, the document is annotated with the CUI under which that term is listed. Ambiguity is introduced by this markup process because the lexical resources often list a particular term as a possible realisation of more than one concept or CUI, as with the *trauma* example above, in which case the document is annotated with all of these possible CUI’s.

The number of tokens of UMLS terms included by this annotation process is given in Table 1. The table shows how many tokens were found by the annotation process, listed according to how many possible senses each of these tokens was assigned in UMLS (so that the number of ambiguous tokens is the number of tokens with more than one possible sense). The greater number of concepts found in the English corpus reflects the fact that UMLS has greater coverage for English than for German, and also that there are many small terms in English which are expressed by single words which would be expressed by larger compound terms in German (for example *knee* + *joint* = *kniegelenk*). Table 1 also shows how many tokens of UMLS concepts were in the annotated corpus *after* we applied the disambiguation process described in Section 4, which proved to be our most successful method. As can be seen, our disambiguation methods resolved some 83% of the ambiguities in the English corpus and 87% of the ambiguities in the German corpus (we refer to this proportion as the ‘Coverage’ of the method). However, this only measures the number of disambiguation decisions that were made: in order to determine how many of these decisions were correct, evaluation corpora were needed.

⁵ <http://link.springer.de/>

Number of Senses	1	2	3	4
Before Disambiguation				
English	223441	31940	3079	56
German	124369	7996	0	0
After Disambiguation				
English	252668	5299	568	5
German	131302	1065	0	0

Table 1

The number of tokens of terms that have 1, 2, 3 and 4 possible senses in the Springer corpus

2.3 Evaluation Corpora

An important aspect of word sense disambiguation is the evaluation of different methods and parameters, some of which have become standardised in the SENSEVAL competitions (Kilgarrieff and Rosenzweig, 2000). Unfortunately, there is still a lack of test sets for evaluation for languages other than English, especially for specific domains like medicine. Given that our work focuses on German as well as English text in the medical domain, we had to develop our own evaluation corpora in order to test our disambiguation methods.

Because in the MUCHMORE project we developed an extensive format for linguistic and semantic annotation (Vintar et al., 2002) that includes annotation with UMLS concepts, we could automatically generate lists of all ambiguous UMLS types (English and German) along with their token frequencies in the corpus. Using these lists we selected a set of 70 frequent types for English (token frequencies at least 28, 41 types having token frequencies over 100). For German, we only selected 24 ambiguous types (token frequencies at least 11, 7 types having token frequencies over 100) because there are fewer ambiguous terms in the German annotation (see Table 1). We automatically selected instances to be annotated using a random selection of occurrences if the token frequency was higher than 100, and using all occurrences if the token frequency was lower than 100. The level of ambiguity for these UMLS terms is mostly limited to 2 senses; only 7 English terms have 3 senses.

Correct senses of the English tokens in context were chosen by three medical experts, two native speakers of German and one of English. The German evaluation corpus was annotated by the two German speakers. Interannotator agreement for individual terms ranged from very low to very high, with an average of 65% for German and 51% for English (where all three annotators agreed). The reasons for this low score are still under investigation. In some cases, the UMLS definitions were insufficient to give a clear distinction between

concepts, especially when the concepts came from different original thesauri. This allowed the decision of whether a particular definition gave a meaningful ‘sense’ to be more or less subjective. Approximately half of the disagreements between annotators occurred with terms where interannotator agreement was less than 10%, which is evidence that a significant amount of the disagreement between annotators was on the *type* level rather than the *token* level. In other cases, it is possible that there was insufficient contextual information provided for annotators to agree. If one of the annotators was unable to choose any of the senses and declared an instance to be ‘unspecified’, this also counted against interannotator agreement. Whatever is responsible, our interannotator agreement fell far short of the 88%-100% achieved in the first SENSEVAL (Kilgarriff and Rosenzweig, 2000, §7), and until this problem is solved or better datasets are found, this poor agreement casts doubt on the generality of the results obtained in this paper.

A ‘gold standard’ was produced for the German UMLS evaluation corpus and used to evaluate the disambiguation of German UMLS concepts. The English experiments were evaluated on those tokens for which the annotators agreed. More details and discussion of the annotation process is available in the project report (Widdows et al., 2003b).

In the rest of this paper we describe the techniques that used these resources to build systems for word sense disambiguation. We compare our results with the evaluation corpora, and investigate the effect of one of the methods in a cross lingual information system.

3 Bilingual Disambiguation

The mapping between word-forms and senses differs across languages, and for this reason the importance of word sense disambiguation has long been recognised for machine translation. By the same token, pairs of translated documents naturally contain information for disambiguation. For example, if in a particular context the English word *drugs* is translated into French as *drogues* rather than *medicaments*, then the English word *drug* is being used to mean *narcotics* rather than *medicines*. This observation has been used for some years on varying scales. Brown et al. (1991) pioneered the use of statistical WSD for translation, building a translation model from one million sentences in English and French. Using this model to help with translation decisions (such as whether *prendre* should be translated as *take* or *make*), the number of acceptable translations produced by their system increased by 8%. Gale et al. (1992) use parallel translations to obtain training and testing data for word-sense disambiguation. Ide (1999) investigates the information made available by a translation of George Orwell’s *Nineteen Eighty-four* into six languages,

using this to analyse the related senses of nine ambiguous English words into hierarchical clusters.

These applications have all been case studies of a handful of particularly interesting words. The large scale of the semantic annotation carried out by the MUCHMORE project has made it possible to extend the bilingual disambiguation technique to entire dictionaries and corpora.

To disambiguate an instance of an ambiguous term, we consulted the translation of the abstract in which it appeared. We regarded the translated abstract as disambiguating the ambiguous term if it met the following two criteria:

- Only one of the CUI’s was assigned to any term in the translated abstract.
- At least one of the terms to which this CUI was assigned in the translated abstract was unambiguous (i.e. was not also assigned another CUI).

3.1 Results for Bilingual Disambiguation

We attempted both to disambiguate terms in the German abstracts using the corresponding English abstracts, and to disambiguate terms in the English abstracts using the corresponding German ones. In this collection of documents, we were able to disambiguate 1802 occurrences of 63 English terms and 1500 occurrences of 43 German terms. Comparing this with the evaluation corpora gave the results in Table 2.⁶

	Precision	Recall	Coverage
English	81%	18%	22%
German	66%	22%	33%

Table 2
Results for bilingual disambiguation

As can be seen, the recall and coverage of this method is not especially good but the precision (at least for English) is very high. The German results contain roughly the same proportion of correct decisions as the English, but many more incorrect ones as well.

⁶ In all of the results presented in this paper, Precision is the proportion of decisions made which were correct according to the evaluation corpora, Recall is the proportion of instances in the evaluation corpora for which a correct decision was made, and Coverage is the proportion of instances in the evaluation corpora for which any decision was made. It follows that

$$\text{Recall} = \text{Precision} \times \text{Coverage}.$$

Our disambiguation results break down into three cases:

- (1) Terms ambiguous in one language that translate as multiple unambiguous terms in the other language; one of the meanings is medical and the other is not.
- (2) Terms ambiguous in one language that translate as multiple unambiguous terms in the other language; both of the terms are medical.
- (3) Terms that are ambiguous between two meanings that are difficult to distinguish.

One striking aspect of the results was that relatively few terms were disambiguated to different senses in different occurrences. This phenomenon was particularly extreme in disambiguating the German terms; of the 43 German terms disambiguated, 42 were assigned the same sense every time we were able to disambiguate them. Only one term, *Metastase*, was assigned different senses; 88 times it was assigned CUI C0027627 (“The spread of cancer from one part of the body to another ...”, associated with the English term *Metastasis* and 6 times it was assigned CUI C0036525 “Used with neoplasms to indicate the secondary location to which the neoplastic process has metastasized”, corresponding to the English terms *metastatic* and *secondary*). *Metastase* therefore falls into category 2 from above, although the distinction between the two meanings is relatively subtle.

The first and third categories above account for the vast majority of cases, in which only one meaning is ever selected. It is easy to see why this would happen in the first category, and it is what we want to happen. For instance, the German term *Krebse* can refer either to *crabs* (Crustaceans) or to *cancerous growths*; it is not surprising that only the latter meaning turns up in the corpus under consideration and that we can determine this from the unambiguous English translation *cancers*.

In English somewhat more terms were disambiguated multiple ways: eight terms were assigned two different senses across their occurrences. All three types of ambiguity were apparent. For instance, the second type (ambiguity between two medical concepts) appeared for the term *Aging*, which can refer either to aging people (*Alte Menschen*) or to the process of aging itself (*Altern*); both meanings appeared in our corpus.

In general, the bilingual method correctly found the meanings of approximately one fifth of the ambiguous terms, and makes only a few mistakes for English but many more for German.

4 Disambiguation using related UMLS terms found in the same context

As well as the specialist lexicon of medical terms in several language, which can be used to disambiguate some terms given a suitable parallel corpus, UMLS gives many relationships between pairs of concepts, as described in Section 2.1. These include traditional thesaural relationships, such as ‘is_a’ (hyponymy) and ‘mapped_to’ (synonymy with a chosen ‘proper form’), which are given by the MRREL and MRCXT files in the UMLS source. There are also lists of ‘coindexing concepts’ (those which have been used by expert medical librarians to index the same MEDLINE abstract), which are given by the MRCOC file in the UMLS source.

These relationships are given between *concepts* (senses of terms) rather than between terms. For each separate sense of an ambiguous word, it is therefore possible to collect a set of related concepts, and if examples of any of these related concepts are found in the corpus near to one of the ambiguous words, it can indicate that the correct sense of the ambiguous word was the one related to this particular concept.

This method is effectively one of the many variants of Lesk’s (1986) original dictionary-based method for disambiguation, where the words appearing in the definitions of different senses of ambiguous words are used to indicate that those senses are being used if they are observed near the ambiguous word. However, we gain over purely dictionary-based methods because the words that occur in dictionary definitions rarely correspond well with those that occur in text. The information we collected from UMLS did not suffer from this drawback: the pairs of coindexing concepts from MRCOC were derived *precisely* from human judgements that these two concepts both occurred in the same text in MEDLINE.

The disambiguation method proceeds as follows. For each ambiguous word w , we find its possible senses $\{s_j(w)\}$. For each sense s_j , find all CUI’s in MRREL, MRCXT or MRCOC files that are related to this sense, and call this set $\{c_{\text{rel}}(s_j)\}$. Then for each occurrence of the ambiguous word w in the corpus we examine the local context to see if a term t occurs whose sense⁷ (CUI) is one of the concepts in $\{c_{\text{rel}}(s_j)\}$, and if so take this as positive evidence that the sense s_j is the appropriate one for this context, by increasing the score of s_j by 1. In this way, each sense s_j in context gets assigned a score which measures the number of terms in this context which are related to this sense. Finally, choose the sense with the highest score.

⁷ This fails to take into account that the term t might itself be ambiguous — it is possible that results could be improved still further by allowing for *mutual* disambiguation of more than one term at once (Stevenson, 2003).

ENGLISH RESULTS	Related terms (MRREL)			Related terms (MRCXT)			Coindexing terms (MRCOC)			Combined (majority voting)		
	Prec	Rec	Cov	Prec	Rec	Cov	Prec	Rec	Cov	Prec	Rec	Cov
Sentence	50	14	28	60	9	15	78	32	41	74	32	43
Document	48	24	50	63	22	35	74	46	62	72	45	63
Subdomain	51	33	65	64	38	59	74	49	66	71	49	69

Table 3

Results for disambiguation based on UMLS relationships (English)

One open question for this algorithm is what region of text to use as a context-window. We experimented with using sentences, documents and whole subdomains, where a ‘subdomain’ was considered to be all of the abstracts appearing in one of the journals in the Springer corpus, such as *Arthroscopie* or *Der Chirurg*. Thus our results (for each language) vary according to which knowledge sources were used (Conceptually Related Terms from MRREL and MRCXT or coindexing terms from MRCOC, or a combination), and according to whether the context-window for recording cooccurrence was a sentence, a document or a subdomain.

4.1 Results for disambiguation based on related UMLS concepts

The results obtained using this method (Tables 3 and 4) were excellent, preserving (and in some cases improving) the high precision of the bilingual and collocational methods while greatly extending coverage and recall. The results obtained by using the coindexing terms for disambiguation were particularly impressive, which coincides with a long-held view in the field that terms which are topically related to a target word can be much richer clues for disambiguation than terms which are (say) hierarchically related. We are very fortunate to have such a wealth of information about the cooccurrence of pairs of concepts through UMLS: the compilation of this information is of course very labour-intensive, and though this labour was not originally for the purpose of word-sense disambiguation, it appears to have provided the benefits of cooccurrence information from manually annotated training data without having to perform any costly manual annotation on a new lexical sample.

In particular, for English (Table 3), results were actually better using only coindexing terms rather than combining this information with hierarchically related terms: both precision and recall are best when using only the MRCOC knowledge source. As we had expected, recall and coverage increased but precision decreased slightly when using larger contexts.

The German results (Table 4) were slightly different, and even more successful, with nearly 60% of the evaluation corpus being correctly disambiguated, nearly 80% of the decisions being correct. Here, there was some small gain when

GERMAN RESULTS	Related terms (MRREL)			Related terms (MRCXT)			Coindexing terms (MRCOC)			Combined (majority voting)		
	Prec	Rec	Cov	Prec	Rec	Cov	Prec	Rec	Cov	Prec	Rec	Cov
Sentence	64	24	38	75	11	15	76	29	38	77	31	40
Document	68	43	63	75	27	36	79	52	66	79	53	67
Subdomain	70	51	73	74	52	70	79	58	73	79	58	73

Table 4

Results for disambiguation based on UMLS relationships (German)

combining the knowledge sources, though the results using only coindexing terms were almost as good. For the German experiments, using larger contexts resulted in greater recall *and* greater precision. This was unexpected — one hypothesis is that the sparser coverage of the German UMLS contributed to less predictable results on the sentence level.

The finding that coindexing terms are more effective clues for disambiguation than hierarchically related terms could be partly a result of the particular UMLS knowledge sources used, rather than a generally valid observation. However, there is mounting evidence that this result *is* part of a more general picture which suggests that syntagmatically (contextually) related terms are at least as useful for disambiguation as paradigmatically (hierarchically) related terms: for example, Patwardhan et al. (2002) discovered that using a Lesk-style gloss-overlap measure of similarity on WordNet was a more successful similarity measure for disambiguation than any of the standard measures of similarity between concepts in the WordNet hierarchy (Budanitsky and Hirst, 2001).

These results are comparable with some of the better SENSEVAL results (Kilgariff and Rosenzweig, 2000) which used fully supervised methods, though the comparison may not be accurate because we are choosing between fewer senses than on average in SENSEVAL, and because of the doubts over our interannotator agreement.

Comparing these results with the number of words disambiguated in the whole corpus (Table 1), it is apparent that the average coverage of this method is actually *higher* for the whole corpus (over 80%) than for the words in the evaluation corpus. It is possible that this reflects the fact the the evaluation corpus was specifically chosen to include words with ‘interesting’ ambiguities, which might include words which are more difficult than average to disambiguate. It is possible that over the whole corpus, the method actually works *even better* than on just the evaluation corpus.

This technique is quite groundbreaking, because it shows that a lexical resource derived almost entirely from English data (MEDLINE indexing terms) could successfully be used for automatic disambiguation in a German corpus. (The alignment of documents and their translations was not even considered

for these experiments so the results do not depend at all on our having access to a parallel corpus.) This is because the UMLS relationships are defined between concepts rather than between words. Thus if we know that there is a relationship between two concepts, we can use that relationship for disambiguation, even if the original evidence for this relationship was derived from information in a different language from the language of the document we are seeking to disambiguate. We are assigning the correct senses based not upon how terms are related in language, but how *medical concepts* are related to one another. It follows that this technique for disambiguation should be applicable to any language which UMLS covers, and applicable at very little cost.

5 Document Retrieval Experiments

The purpose of the semantic annotation and disambiguation of the Springer corpus was to develop techniques for multilingual access to medical documents, and the contribution of these annotation and disambiguation components can be measured to some extent by their effect on precision and recall in document retrieval experiments. Such experiments were carried out, comparing the performance of a cross lingual search engine which made use of the semantic annotation and disambiguation, with a baseline search engine which relied only on the surface forms (tokens) appearing in the documents.

The test sets used for evaluation, consisting of 25 specially designed queries for each language and relevance judgements from medical experts, are described by (Volk et al., 2002). The queries were translated into both languages, so that as well as monolingual retrieval runs, the system could be evaluated on its performance cross lingually, matching queries in German to documents in English and *vice versa*.

The search engine used in these experiments was a cross lingual vector model compiled from the parallel corpus. The model uses latent semantic analysis, combining the techniques used by Schütze (1998) for word sense discrimination and Littman et al. (1998) for CLIR: the creation of such a bilingual vector model from the Springer corpus is described in more detail in (Widdows et al., 2002).

Three search engines using this vector model were built, which indexed each document using

- Tokens only
- UMLS concepts (CUIs) (automatically added, no disambiguation)
- UMLS concepts, disambiguated using the method of Section 4.

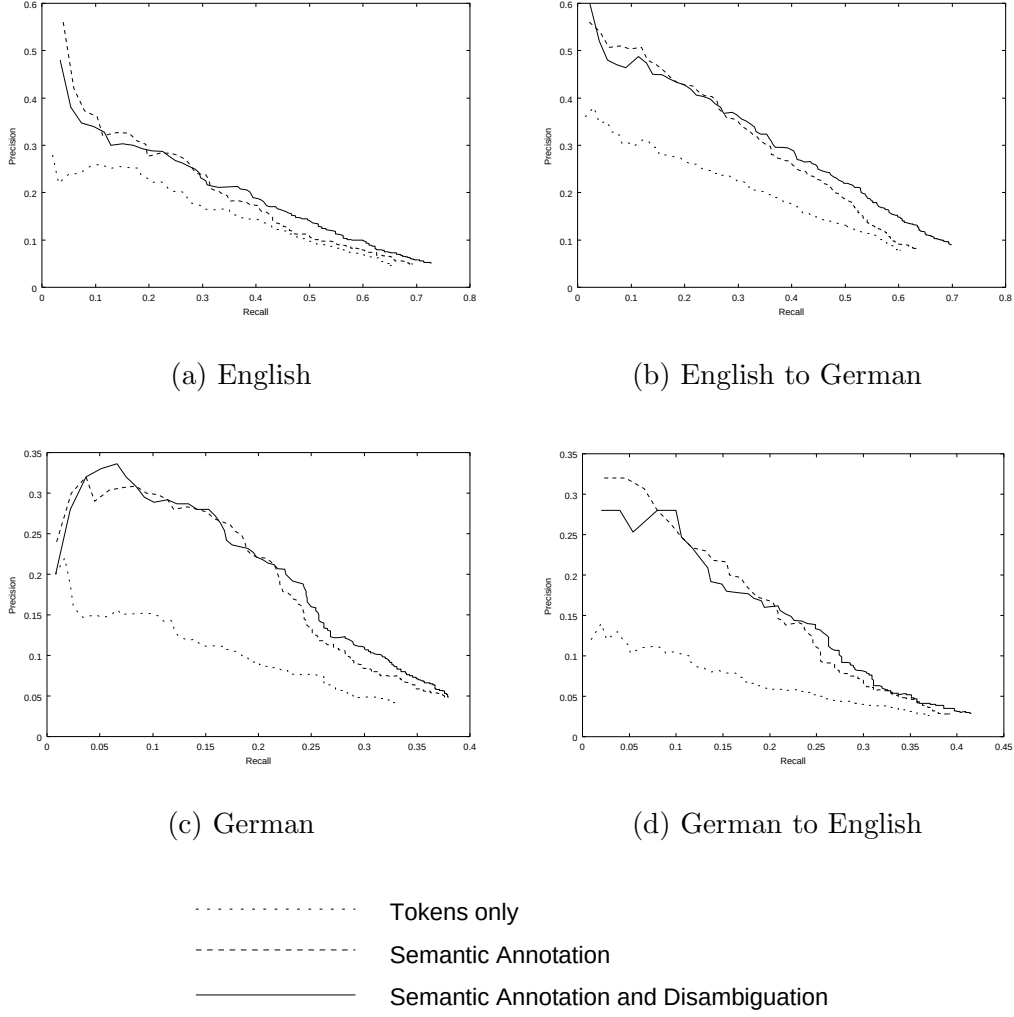


Fig. 1. Precision / Recall curves for different retrieval experiments

The search engine built using automatic disambiguation applied to technique of Section 4 where possible to disambiguate both the queries and the documents.

For each search engine, four retrieval tasks were carried out (English, German, English queries to German documents, German queries to English documents), giving a total of 12 experiments. Precision and recall for the documents retrieved in each of these experiments are shown in Figure 1. The Mean Average Precision obtained in each experiment (a single number measuring the average precision achieved after each document is retrieved, for the top 200 retrieved documents) is given in Table 5.

For each of the four retrieval tasks, the search engines which used the semantic annotation (with or without disambiguation) performed significantly better than the baseline model which used tokens only. This disparity was much

	English	German	English to German	German to English
Tokens only	0.100	0.079	0.152	0.050
Semantic Annotation	0.111	0.110	0.184	0.071
Semantic Annotation and Disambiguation	0.115	0.113	0.193	0.071

Table 5

Mean Average Precision for each experiment

greater for German than for English. We believe that this is mainly because the ‘tokens only’ model is much worse for German, rather than because the semantic annotation is better: this hypothesis is consistent with the findings of (Volk et al., 2002, §4.2), who show that compounding and stemming terms (so that documents can be represented by their lemmas rather than simply their tokens) improves performance for German but has a mainly negative effect for English.

Adding the disambiguation component to the system gave reliable, though slight, improvements over the search engine built using semantic annotation but not disambiguation, for each of the experiments except for the English queries to German documents run. The automatic disambiguation improves the precision of high-recall retrieval, as can be seen from the curves in Figure 1. However, high-precision retrieval (the relevance of the top few documents returned at the top of the list by the search engine) was *not* improved by disambiguation, and (especially with English queries) it could actually be slightly harmed. The reasons for this difference between high-precision and high-recall retrieval are as yet unclear.

The potential importance of word sense disambiguation to improving intelligent information retrieval has long been recognised, though in practice it has proved tremendously difficult to get any demonstrable improvement: experimental evidence so far is that automatic disambiguation has more often harmed than helped retrieval (Sanderson, 2000). This finding that has encouraged researchers to look very closely at different methodological issues to try to establish principles and causes behind good and bad results (Krovetz, 2002). Because of this, the marginal improvement in retrieval performance gained by our disambiguation system is quite encouraging: having used only a freely available language resource to give relationships between concepts, the disambiguation system gives a reliable improvement in retrieval performance without using any manually annotated training data.

6 Conclusions

The goal of this work was to develop techniques which could be used to disambiguate all of the ambiguous medical terms in a corpus of both English and German documents, without relying on any manually annotated training examples (which could only have been produced for a few selected terms).

This paper demonstrates that instead of manually annotated training data, a multilingual language resource such as UMLS can be used for both cross lingual and monolingual word sense disambiguation. Our most successful and encouraging result is the discovery that relationships between medical concepts, mined from the index of an *English* library, can be used automatically to disambiguate *German* documents with high precision and coverage, at no extra cost.

Using this disambiguation technique as a component in a multilingual information system gave slight but reliable improvements in the precision of high-recall document retrieval.

Our experiments demonstrate that knowledge-rich information, such as cooccurrence in a library catalogue, can provide excellent models of the behaviour of and relationships between different concepts. Such models are a viable alternative to manually annotated training data, and can be much more readily scaled up to disambiguate the important terms in a whole document collection.

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